

Budget-Constrained Aid Allocation as a POMDP

1st Xinyue Zhang

Department of Computer Science
Stanford University
Stanford, USA
zhangsx@stanford.edu

2nd Abi Lopez

Department of Computer Science
Stanford University
Stanford, USA
abilopez@stanford.edu

Abstract—Allocating limited food and housing assistance to financially unstable households is a challenging decision-making problem due to uncertainty in both observable need and latent household resilience. We propose a data-driven simulation and planning framework that models each family’s food and housing stability as a partially observable Markov decision process (POMDP). The simulator incorporates empirically grounded household cost burdens, heterogeneous baseline income coverage, slow-moving housing dynamics, and noisy observations. From simulated trajectories, we learn a compact single-family POMDP and apply point-based value iteration (PBVI) to derive belief-based policies that maximize the reward across all families. We evaluate PBVI and compare to several baselines (greedy, random, myopic POMDP, and spread policy variations) within a multi-family environment subject to a binding budget constraint. Results show that in medium budget and longer horizon cases, PBVI improves cumulative reward over feasible baselines and exhibits temporally coherent aid patterns, but crisis rates may remain high across budget ranges due to structural fragility encoded in the learned transition dynamics. Our findings highlight both the promise and limitations of POMDP-based allocation strategies in welfare settings characterized by uncertainty and persistent instability.

Index Terms—Partially observable Markov decision processes, welfare allocation, food insecurity, housing instability, stochastic planning, PBVI, sequential decision-making.

I. INTRODUCTION

Food and housing stability are critical determinants of well-being, yet millions of U.S. households routinely face hardship due to income volatility, structural inequality, and uneven access to public benefits. Households may move rapidly from stability to hardship or crisis as their trajectories evolve in nonlinear and highly volatile ways. For organizations that distribute emergency assistance, a central challenge is determining how to allocate limited resources between many families, each with different needs, risk profiles, and hidden sources of fragility.

These indicators reveal surface conditions but not the hidden factors that govern stability, such as informal income, debt, or social support. Two households with similar reported income may differ drastically in resilience, and the effect of assistance varies accordingly: a small food voucher might fully stabilize one family but barely help another. Housing interventions are especially difficult, as small subsidies rarely prevent eviction or chronic rent burden.

Previous research in the field has documented these issues and their severity. In 2023, households in the lowest income

quintile spent over 32% of their after-tax income on food alone [1], and nearly half of U.S. renter households were cost-burdened [2]. Housing markets exhibit persistent shortages of affordable units, and millions of households meet HUD’s definition of “worst-case housing needs” [3]. Meanwhile, food insecurity remains widespread, and evidence shows that nutrition assistance programs exhibit substantial regional variation in adequacy, compounding household risk [4].

Despite this complexity, algorithmic approaches to resource allocation often assume full observability of household conditions or deterministic transitions [5]–[7]. These assumptions contradict decades of policy research showing that instability is driven by latent factors such as informal labor income, interpersonal obligations, health shocks, and fluctuating subsidies.

To bridge this gap, we frame aid allocation as a sequential decision-making problem under partial observability. We model households as evolving in discrete stability states and assume the planner receives only noisy ordinal observations. This perspective allows for belief-based decision-making, where the system maintains probabilistic estimates of latent household fragility rather than relying solely on surface-level indicators.

We make three primary contributions:

- 1) We introduce a household instability simulator that incorporates income-based heterogeneity, realistic cost structures, asymmetric recovery speeds, and noisy observations.
- 2) We learn a compact single-family POMDP from simulated trajectories and solve it using point-based value iteration (PBVI), producing a belief-based decision rule for aid allocation.
- 3) We evaluate the resulting policies in a multi-family environment with binding budget constraints and analyze performance across horizons, budgets, and household counts.

II. RELATED WORK

Resource allocation under uncertainty has been widely studied in operations research and decision-theoretic planning. Classical formulations apply stochastic optimization to constrained resource deployment where demand evolves over time [8]. More recent work models emergency response and medical triage problems as partially observable Markov decision processes (POMDPs), where decision-makers lack full

visibility into individual risk states [9], [10]. These studies emphasize that optimal triage must account not only for immediate need but also for how interventions affect long-term risk trajectories—a perspective directly relevant to poverty alleviation and household stabilization.

Research in economics and housing policy highlights the multidimensional nature of basic welfare: food and housing insecurity jointly predict long-term financial instability. Low-income households face high food expenditures [1], widespread rent burden and affordability shortages [2], [11], and heterogeneous benefit adequacy [4]. These structural factors generate instability even at moderate rent burdens [12] and motivate models that go beyond observable income when allocating support.

Similarly, studies of food assistance and rent burden show substantial heterogeneity in how support translates into stability. SNAP benefits rarely cover full nutritional requirements for most households, and local variability in food prices magnifies differences in adequacy [4]. Even rent burdens below 50% can still propagate instability due to income volatility and competing expenses [12]. However, despite this evidence of hidden financial heterogeneity, prior algorithmic literature typically assumes directly observable socioeconomic states when optimizing public resource allocation [5]–[7].

Our work addresses this gap by explicitly modeling latent household resilience using hidden baseline income parameters, combined with noisy observational signals. This structure enables belief-based policies (PBVI) to infer which families are inherently unstable and prioritize aid accordingly, without direct access to income or support levels. By embedding empirically grounded asymmetries in state transitions—where food insecurity responds more quickly to aid than housing instability [1], [3]—the proposed simulator provides a principled and realistic testbed for evaluating algorithmic welfare allocation under uncertainty.

III. METHODOLOGY

Modeling food and housing instability requires simplifying socioeconomic processes that are difficult to observe directly. Because planners also lack access to latent variables such as informal income or debt, our goal is to approximate the decision environment under partial observability rather than replicate every causal mechanism.

We use a compressed model that captures the evolution of food and housing stability while embedding deeper financial heterogeneity implicitly through stochastic transition dynamics. The parameters we adopt are empirically grounded but intentionally simplified to balance realistic behavior and tractable belief-based planning. Importantly, these choices influence what the planner learns; in our discussion, we examine how our assumptions shape PBVI performance and how varying them could alter allocation strategies.

A. Formal Problem Definition

We model multi-family aid allocation as a finite-horizon partially observable Markov decision process (POMDP)

$$\mathcal{M} = \langle S, \mathcal{A}, \mathcal{O}, T, Z, R, H, B \rangle,$$

where S is the latent household stability space, $\mathcal{A}$ is the discrete aid-allocation space, $\mathcal{O}$ is the set of noisy ordinal observations, T is the transition model, Z is the observation likelihood model linking latent states to observed reports, and R is a humanitarian reward function. The parameter H denotes the planning horizon in months, and B is the shared dollar budget that constrains the total aid allocated across all families. The simulator tracks additional latent household characteristics (including income-based baseline coverage), but the planning model does not operate on these quantities directly. Instead, the POMDP represents a compressed abstraction whose transition dynamics are statistically learned from noisy observations generated by the simulator. Thus, the planner reasons only over stability levels (f, h) while treating underlying financial resilience as hidden structure revealed implicitly through observational data.

a) *State Space*: Each family i has a hidden state

$$x = (f, h) \in \{0, 1, 2\}^2, \quad (1)$$

where f and h denote food and housing need levels (Crisis = 0, Moderate Need = 1, Stable = 2). Income-dependent baseline resilience is represented implicitly in the simulator (Sec. II-B) and is not directly observed by the planner.

b) *Action Space*: The planner allocates discrete aid units:

$$a = (u_f, u_h), \quad u_f \in \{0, 1, 2\}, \quad u_h \in \{0, 1, 2, 3, 4, 5\}, \quad (2)$$

where each unit corresponds to \$500 of support. The planner begins each episode with a fixed total budget, and every allocated aid unit deducts its cost from the remaining amount. Therefore, the feasible action set is not fixed but shrinks dynamically with remaining resources. At time t , the agent selects from a restricted subset $\mathcal{A}_t \subseteq \mathcal{A}$ such that the total cost of any joint allocation does not exceed the current available budget.

c) *Observation Space*: The planner receives noisy ordinal observations

$$o = (\tilde{f}, \tilde{h}) \in \{0, 1, 2\}^2, \quad (3)$$

generated from a discrete Gaussian centered at the true state. Together with the transition model, the observation likelihood $Z(o \mid x')$ drives Bayesian belief updates over latent states. Although the planner never observes baseline income directly, its effect on recovery trajectories is implicitly inferred through these belief updates.

d) *Transition Model*: The joint transition factors across food and housing:

$$T(x \rightarrow x' \mid a) = T_{u_f}(\text{food})(f \rightarrow f') \cdot T_{u_h}(\text{housing})(h \rightarrow h'),$$

where each T_{u_f} and T_{u_h} is a stochastic kernel whose probabilities are parameterized by an aid-induced *closure fraction*

that reflects the portion of unmet need covered by income and assistance (defined in Sec. II-B). This embeds latent financial resilience into the transition dynamics without exposing it directly to the planner.

e) Reward Function: We use a humanitarian objective penalizing crisis states:

$$R(x', a) = w_f f' + w_h h' - \lambda [\mathbf{1}(f' = 0) + \mathbf{1}(h' = 0)], \quad (4)$$

encoding that preventing crisis is more critical than incremental improvement.

f) Objective: The planner seeks a policy π maximizing expected total reward subject to the budget:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=1}^H R(x'_t, a_t) \right] \quad \text{s.t.} \quad \sum_{t=1}^H \sum_i \text{cost}(a_t^{(i)}) \leq B. \quad (5)$$

IV. SIMULATOR AND POMDP ABSTRACTION

To model real-world uncertainty without revealing latent household attributes to the planner, we generate synthetic trajectories using a stochastic simulator with hidden income-based resilience parameters. These latent baselines influence how families recover from crisis but are never observed directly; instead, the planner only receives noisy ordinal signals of food and housing stability. By injecting realistic observation noise and forbidding access to latent parameters, the simulator avoids data leakage and yields a learning environment where transition models must be inferred statistically rather than read from ground truth. This enables a faithful POMDP abstraction in which hidden resilience is represented implicitly through closure-dependent transition dynamics rather than explicit state features.

Each episode simulates N families over a horizon of H months, with a shared dollar budget B . For family i at time t , the hidden state is

$$x_t^{(i)} = (\text{food}_t^{(i)}, \text{housing}_t^{(i)}, \alpha_{\text{food}}^{(i)}, \alpha_{\text{housing}}^{(i)}), \quad (6)$$

where $\text{food}, \text{housing} \in \{0, 1, 2\}$ represent crisis (0), moderate need (1), and stable (2) levels. The parameters $\alpha_{\text{food}}^{(i)}, \alpha_{\text{housing}}^{(i)} \in [0, 1]$ represent fixed baseline coverage fractions from income, subsidies, and informal support.

We approximate baseline distributions as

$$\alpha_{\text{food}}^{(i)} \sim \text{Beta}(6, 3) \quad (E \approx 0.67), \quad (7)$$

$$\alpha_{\text{housing}}^{(i)} \sim \text{Beta}(3, 7) \quad (E \approx 0.30), \quad (8)$$

reflecting that low-income households tend to meet a majority of food needs but substantially less of housing needs without assistance [1]–[3].

At each decision epoch, the planner selects an aid action for each family:

$$a_t^{(i)} = (u_{\text{food}}^{(i)}, u_{\text{housing}}^{(i)}), \quad (9)$$

where each unit corresponds to \$500 of support. Food and housing units are capped at (2, 5) per month to reflect plausible aid limits.

a) Closure Fraction: Aid units are translated into a closure fraction:

$$\phi_{\text{food}}^{(i)} = \min \left(1, \alpha_{\text{food}}^{(i)} + \frac{500 u_{\text{food}}^{(i)}}{\text{FOOD_TARGET}} \right), \quad (10)$$

$$\phi_{\text{housing}}^{(i)} = \min \left(1, \alpha_{\text{housing}}^{(i)} + \frac{500 u_{\text{housing}}^{(i)}}{\text{HOUSING_TARGET}} \right), \quad (11)$$

with $\text{FOOD_TARGET} = \$1000$ and $\text{HOUSING_TARGET} = \2500 . These target values are informed by cost of living estimates from the MIT Living Wage Calculator for a California household with two adults (one working) and two children, which reports monthly costs of approximately \$1,115 for food and \$2,500 for housing [13].

b) Transition Model: For each dimension $d \in \{\text{food}, \text{housing}\}$, the transition from level $k \in \{0, 1, 2\}$ depends on ϕ_d . For example, for food:

$$\Pr(0 \rightarrow 1) = \text{clip}(0.10 + 0.60\phi_{\text{food}}, 0, 1), \quad (12)$$

$$\Pr(1 \rightarrow 2) = \text{clip}(0.05 + 0.40\phi_{\text{food}}, 0, 1), \quad (13)$$

$$\Pr(1 \rightarrow 0) = \text{clip}(0.20(1 - 0.50\phi_{\text{food}}), 0, 1), \quad (14)$$

$$\Pr(2 \rightarrow 1) = 0.08(1 - 0.70\phi_{\text{food}}). \quad (15)$$

Housing uses analogous parameters but with slower improvement and faster regression, consistent with rent-burden and eviction dynamics [3], [11]. Assuming conditional independence, the joint transition factorizes across food and housing.

c) Observations: The planner observes noisy need levels:

$$o_t^{(i)} = (\tilde{\text{food}}_t^{(i)}, \tilde{\text{housing}}_t^{(i)}), \quad (16)$$

where each component is drawn from a discrete Gaussian centered at the true level with variances $\sigma_{\text{food}}^2 = 0.8^2$ and $\sigma_{\text{housing}}^2 = 0.6^2$. Baseline income α remains hidden.

d) Reward: The per-family reward is

$$r(x_t^{(i)}) = w_f \cdot \text{food}_t^{(i)} + w_h \cdot \text{housing}_t^{(i)} \quad (17)$$

$$- \lambda [\mathbf{1}(\text{food}_t^{(i)} = 0) + \mathbf{1}(\text{housing}_t^{(i)} = 0)] \quad (18)$$

with simulator parameters $(w_f, w_h, \lambda) = (2.0, 3.0, 6.0)$.

A. Dataset Generation

Trajectories are generated by simulating an epsilon-greedy exploration policy. Random actions ensure broad coverage of the state-action space, while targeted actions prioritize crisis-level households. A budget-cap heuristic spreads remaining spending across future time steps to avoid degenerate all-at-once allocation.

Each trajectory stores the assigned actions, noisy observations, hidden next states (including α), and realized reward.

B. Learned Single-Family POMDP

From the trajectories we estimate a compact 9-state POMDP over $(f, h) \in \{0, 1, 2\}^2$. Hidden baselines α are not included in the planning state space; their influence enters implicitly through learned transition probabilities.

We estimate:

- priors $p(f_0)$ and $p(h_0)$ via smoothed frequency counts,
- per-unit transition matrices $T_u(d)$ for $d \in \{\text{food}, \text{housing}\}$ and unit count u ,
- observation likelihoods $O_d(o \mid k)$ from aligned noisy observations.

The joint transition is:

$$T((f, h) \rightarrow (f', h') \mid a) = T_{a_f}(\text{food})(f \rightarrow f') \quad (19)$$

$$\cdot T_{a_h}(\text{housing})(h \rightarrow h') \quad (20)$$

with a similar factorization for observations.

The planning reward strengthens crisis penalties:

$$R((f', h'), a) = 5.0 f' + 7.5 h' - 20.0 [1(f' = 0) + 1(h' = 0)]. \quad (21)$$

C. Planning Algorithms

We compare the following policies:

- **PBVI.** Point-based value iteration using 100 belief samples, discount $\gamma = 0.95$, and 8 backup iterations. At deployment, each family updates its belief and receives the feasible action maximizing expected value. Families are sorted by value-per-dollar and allocated greedily under the remaining budget.
- **Myopic POMDP.** Uses belief updates but selects actions maximizing expected immediate reward from the most likely state.
- **Greedy Observation-Based.** Ignores beliefs; ranks families by observed need $(2 - f) + (2 - h)$ and allocates up to one food and one housing unit per high-need family.
- **Random.** Samples feasible units uniformly given the remaining budget.
- **Spread Policies.** “Random spread” and “greedy spread” allocate aid with per-step cap limits for how many units may be assigned in that round. The spread policy then distributes units across families until this capped unit allowance is exhausted.

D. Diagnostics and Testing Framework

We implemented a focused set of diagnostics to verify that the simulator, learned POMDP, and PBVI solver behaved as intended. Basic structural checks ensured that learned transition matrices were valid probability distributions and exhibited the correct qualitative behavior—higher aid yielding higher recovery probabilities and lower regression risk. We also confirmed that belief updates produced well-formed posterior distributions and reacted smoothly to observation noise, resolving a small normalization bug early in development.

More importantly, we conducted end-to-end rollouts under PBVI, greedy, myopic, and random policies to assess whether the overall system produced realistic dynamics. These tests validated that food and housing trajectories matched domain expectations, crisis rates remained within plausible ranges, and the reward function accumulated correctly over time. They also confirmed that multi-family budget logic interacted correctly with per-family action recommendations, preventing infeasible or front-loaded spending.

V. EXPERIMENTS AND RESULTS

We evaluate the learned single-family POMDP, the PBVI-derived policy, and baseline heuristics using (1) simulator and transition validation, (2) multi-episode policy comparisons, and (3) budget sensitivity analyses. Across these settings, our results show that PBVI consistently produces smoother, more structured allocation patterns and higher cumulative reward than simple heuristics; however, even optimal planning struggles to prevent widespread instability when households face slow-moving dynamics—particularly in housing. These findings suggest that algorithmic allocation is limited less by model quality than by structural constraints encoded in the environment.

A. Simulator and Transition Diagnostics

We begin by validating that the simulator and learned POMDP provide a reliable environment for policy evaluation. Our diagnostics examine three components: how different baseline policies affect crisis rates, rewards, and budget use; whether the learned transition model reproduces the intended food–housing dynamics under fixed aid schedules; and how posterior beliefs evolve when exposed to noisy observations. Together, these checks verify that the transition structure, uncertainty model, and belief updates behave coherently, ensuring that subsequent policy comparisons rest on a well-calibrated decision process.

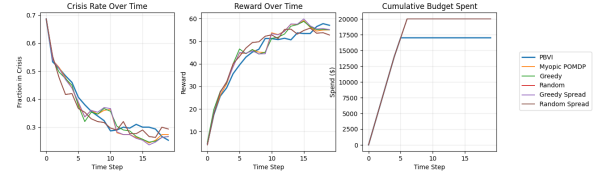


Fig. 1. Crisis rate, reward, and cumulative budget spent over time for six policies: PBVI, Myopic POMDP, Greedy, Random, Greedy Spread, and Random Spread.

Our first set of experiments verifies that the simulator, learned transition model, and evaluation loop behave as intended. Figure 1 reports crisis reduction, reward growth, and cumulative budget expenditure for all six policies. PBVI and the myopic POMDP baseline consistently achieve lower crisis rates than the heuristic and random policies, while also producing smoother reward trajectories. The cumulative-spend curves confirm that the PBVI and greedy families front-load their allocations, whereas random and spread heuristics deplete their budgets immediately. These trends indicate that the learned dynamics support forward-looking planning—improvements materialize gradually, especially in housing—and cannot be forced through naïve early overspending, matching the structure we designed into the system.

To confirm this structure, Figures 2 show that the POMDP transition dynamics behave as expected under repeated fixed actions. The trajectories reproduce the intended asymmetry between the two subsystems: food levels recover quickly under modest aid, while housing responds slowly and remains prone to regression even with stronger assistance. The no-aid baseline remains in persistent crisis for both dimensions,

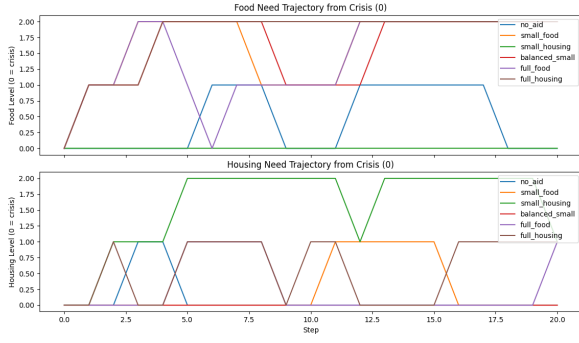


Fig. 2. Food-need trajectory from an initial crisis state under different fixed-aid schedules.

confirming that improvement is driven by intervention rather than stochastic drift. These patterns indicate that the estimated transition matrices are behaving correctly and that the qualitative structure of the simulator—rapid food recovery and sluggish housing recovery—is preserved in the learned model.

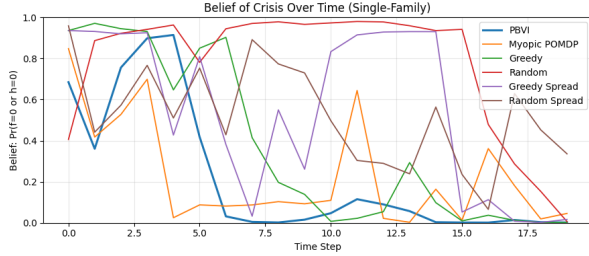


Fig. 3. Belief trajectories of being in crisis for a single family under all six policies. Each curve shows the posterior probability that the latent state satisfies ($f = 0 \vee h = 0$) after updating on noisy observations over time.

Figure 3 shows how all six policies update the belief of being in crisis for a single representative family. PBVI consistently drives the crisis belief toward zero and maintains stability across the horizon, demonstrating robust long-term reasoning even under observation noise. Myopic POMDP and greedy policies reduce crisis belief but exhibit substantial oscillations, reflecting their limited ability to smooth over uncertainty or anticipate delayed effects of aid. Random and spread-based heuristics produce highly volatile trajectories with extended periods of miscalibrated crisis belief, emphasizing their lack of structured state inference. Overall, PBVI yields the most stable and coherent posterior evolution, complementing its stronger performance in the time-series outcome metrics.

B. Multi-Episode Policy Comparison

After validating the simulator and transition structure, we evaluate how different allocation strategies perform when deployed across multiple families with a shared budget. The objective of this experiment is to assess *policy behavior over full episodes*, where early choices compound through delayed food and housing recovery.

We compare PBVI, myopic-POMDP, greedy, random, and spread-based heuristics across 40 episodes using a \$50,000 budget, 20-step horizon, and 8 families. Figure 4 summarizes the resulting episode rewards.

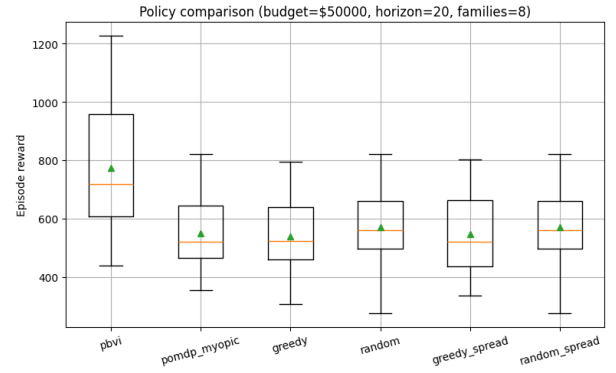


Fig. 4. Episode reward distribution for all policies (budget=\$50k, horizon=20, families=8).

PBVI achieves the highest median reward and the widest upper tail, indicating that belief-based planning can extract sustained benefits from early stabilization. This reflects PBVI’s ability to use belief updates and long-horizon structure to concentrate aid where it yields the most persistent improvements. The myopic-POMDP policy performs moderately but remains well below PBVI, highlighting the limits of short-term optimization in this domain. Greedy, random, and spread-based strategies cluster closely together with lower and more tightly bounded reward distributions, indicating that heuristics lacking state-awareness or temporal planning fail to capture the structural advantages present in the learned dynamics.

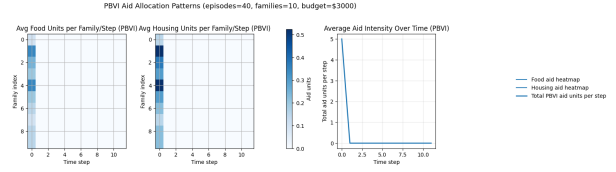


Fig. 5. Aggregated PBVI food and housing allocations across 40 simulated episodes, showing average units delivered per family per time step and the corresponding aid-intensity profile.

Figure 5 reveals a clear and interpretable structure in PBVI’s learned allocation strategy when averaged over many episodes. Rather than distributing aid uniformly or reactively, PBVI consistently front-loads its spending, directing both food and housing units to a small subset of families during the earliest time steps. The aid-intensity curve confirms this deliberate design: PBVI rapidly expends its resources at the start of the horizon and then confidently commits to a zero-allocation regime once acute instability is resolved. This behavior suggests that PBVI has internalized a high-leverage intervention principle—early stabilization yields the greatest long-term payoff under the learned transition dynamics. By uncovering this systematic pattern across episodes, the visualization demonstrates that PBVI is not merely performing greedy triage; it is executing a coherent temporal strategy that exploits the modeled recovery structure, highlighting the promise of POMDP-based planning for understanding and optimizing social-resource allocation.

C. Budget Sensitivity: Crisis-Rate Control

To evaluate robustness under resource constraints, we sweep budgets from \$5k–\$100k across horizons $\{8, 12, 16\}$ and family

counts $\{5, 10, 15\}$, and record the minimum budget at which PBVI achieves a mean crisis rate ≤ 0.4 . Figure 6 reports these thresholds.

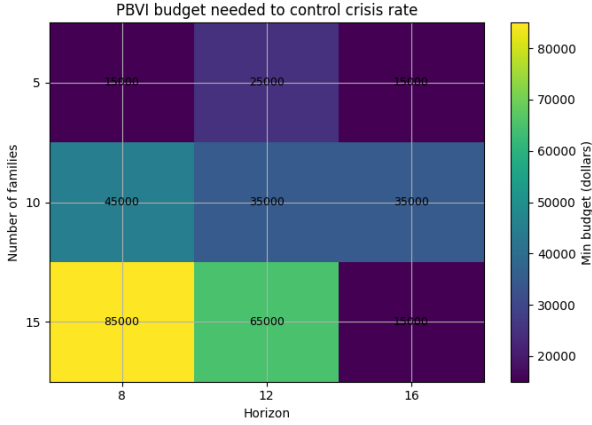


Fig. 6. Minimum PBVI budget required to maintain mean crisis ≤ 0.4 across horizons and family sizes.

The heatmap reveals substantial variation in budget requirements across configurations. For small family sets (5 families), PBVI controls crisis with relatively modest resources (\$15k–\$25k). As the number of families increases to 10 or 15, the required budget typically rises sharply—reaching as high as \$85k for the most difficult setting (15 families, horizon 8). However, longer horizons consistently reduce the needed budget, with the most favorable configuration (15 families, horizon 16) requiring only \$15k. These patterns suggest that PBVI relies heavily on extended planning windows to exploit delayed recovery dynamics; short horizons force more immediate and costly interventions, while longer horizons enable gradual stabilization at significantly lower cost.

VI. DISCUSSION

The results show that PBVI’s performance gains stem from how belief-based planning exploits the structural dynamics of the learned model rather than from optimization alone. PBVI consistently takes advantage of two embedded mechanisms: (1) early leverage, where modest front-loaded aid reshapes long-run trajectories for a subset of households, and (2) temporal asymmetry, where food and housing recover at different speeds. Belief updates allow PBVI to detect households whose slow housing recovery creates persistent risk and to intervene before instability compounds. Heuristics, which respond only to observed need, cannot match this ability to target latent fragility and often misallocate aid or overspend on short-term signals.

At the same time, the model reveals clear limits. When housing recovery is extremely slow or volatile, no plausible budget reliably reduces crisis rates—especially with short horizons or large family sets. In these settings, the barrier is not the policy but the environment itself: recovery pathways are too weak to overcome structural scarcity. PBVI helps identify when improved planning meaningfully shifts outcomes and when deeper socioeconomic constraints dominate. This echoes

real-world welfare contexts, where algorithmic allocation cannot substitute for the broader rights-based conditions—stable housing, livable wages, and access to essential services—that enable families to recover at all.

These findings suggest several avenues for future work. Richer transition models could capture heterogeneous recovery profiles, allowing more tailored intervention strategies. Dynamic budgets or adaptive action spaces may help planning respond to sudden instability. Most importantly, grounding the model in administrative or survey data would clarify which constraints reflect real household hardship and which arise from our simplified simulator. Used in this diagnostic way, PBVI offers insight into both the potential and the limits of algorithmic decision-making in welfare allocation under persistent uncertainty.

VII. CONCLUSION

This work develops a welfare-allocation framework that integrates partial observability, empirically informed instability, and strict budget limits into a unified planning model. By learning a compact single-family POMDP from simulated trajectories and solving it with PBVI, we show that belief-based planning can outperform standard heuristics—especially in environments where housing recovery is slow and structurally fragile. The resulting policies display interpretable patterns such as targeted early intervention and differentiated use of food and housing aid, demonstrating that even a compressed model can capture meaningful allocation structure.

Our hybrid approach—using PBVI to evaluate households and a simple budget-aware selection rule to distribute aid—balances tractability with long-horizon reasoning. It consistently surpasses observation-based and myopic baselines, highlighting the value of uncertainty-aware prioritization in multi-family settings. Yet the results also reveal boundaries that algorithmic refinements alone cannot overcome. When the environment encodes deep structural instability, even generous budgets and sophisticated planning struggle to suppress crisis rates. This reflects a broader truth: planning tools can improve allocation, but long-term stability ultimately depends on conditions beyond the model’s control, such as access to stable housing, adequate income, and supportive public infrastructure.

Viewed together, these findings position the POMDP–PBVI framework not as a prescriptive solution but as a lens for understanding where targeted planning can shift outcomes and where structural factors dominate. By making these mechanisms explicit, the framework helps map the space of feasible improvements and highlights how richer models, better data, and tighter integration with real policy environments could produce even greater gains. In this sense, the work underscores both the potential and the opportunity for continued innovation in algorithmic approaches to household stability.

VIII. DIVISION OF WORK

Xinyue (Shirley) Zhang

She wrote most of the project proposal and status update, and made edits to the writing of the final report. She contributed extensively to designing the POMDP formulation and wrote the code for the data generator used to produce random rollout trajectories that informed the learned transition model. She also wrote the code used to extract and capture the model dynamics from these rollouts. In addition, she implemented the PBVI solver and baseline policies (random, greedy, myopic POMDP, and spread strategies), as well as portions of the testing framework to generate the parameter grid heatmap and the policy comparison box-plot. She also spent considerable time on the underlying components that informed those results, most notably the PBVI backup and the POMDP transition model.

Abi Lopez

She contributed across all stages of the project, with a focus on ensuring the POMDP framework behaved correctly and could be reliably evaluated. She led the development and debugging of the initial exact value-iteration attempt, diagnosed the numerical issues, and helped guide the shift toward PBVI. She also refined parts of the POMDP and PBVI implementations—including belief updates, budgeting logic, and model-policy interfaces—to improve consistency with the learned dynamics. In addition, she developed much of the diagnostic and testing workflow, creating rollout analyzers, belief-evolution checks, and evaluation plots (aid-allocation heatmaps, crisis and reward trajectories, budget-feasibility visualizations) used to validate model behavior. Taking the course for four units, she contributed meaningfully through additional hours spent refining the POMDP and PBVI code, iterating on the evaluation workflow, generating the complete set of plots, and extensively polishing the final report’s analysis. She wrote the experiments and results, discussion, and conclusion sections of the final report, integrating the empirical findings into a clear narrative.

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